



AI-Powered Threat Intelligence Platform

Technical Documentation & Architecture Guide

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Component	Technology
Cloud Platform	Amazon Web Services (AWS)
AI Model	Amazon Nova Lite (Bedrock)
Compute	AWS Lambda (Serverless)
Storage	DynamoDB + S3
Security	VPC + Secrets Manager + IAM
Threat Feeds	AlienVault OTX + AbuseIPDB
Enrichment	VirusTotal + MITRE ATT&CK
Delivery	Email (SES) + Web Dashboard + Telegram

1. Executive Summary

The AI-Powered Threat Intelligence Platform is an enterprise-grade, serverless cybersecurity solution built on Amazon Web Services. It automates the collection, enrichment, analysis, and reporting of cyber threat indicators (IOCs) using artificial intelligence.

The platform addresses a critical operational challenge faced by Security Operations Centers (SOCs): the manual, time-consuming process of gathering and analyzing threat intelligence from multiple public feeds. By automating this workflow, the platform reduces analysis time from 3-4 hours to under 5 minutes while improving coverage and accuracy.

1.1 Key Capabilities

- Automated IOC collection from AlienVault OTX and AbuseIPDB (99+ indicators/day)
- VirusTotal enrichment for confirmation of malicious indicators
- MITRE ATT&CK framework mapping for threat categorization
- AI-powered analysis using Amazon Nova Lite (Bedrock)
- Professional PDF report generation with charts and tables
- Multi-channel delivery: Email (AWS SES) + Web Dashboard + Telegram
- Enterprise security: VPC isolation, Secrets Manager, IAM least privilege

1.2 Business Value

Metric	Without Platform	With Platform
Analysis Time	3-4 hours/day	< 5 minutes
IOC Coverage	20-30 manual	99+ automated
Threat Enrichment	Manual lookup	Automated VT + MITRE
Report Generation	1-2 hours	Automatic PDF
Cost	\$50,000+/year	~\$0.50/month
Availability	Business hours only	24/7 automated

2. System Architecture

The platform follows a serverless, event-driven architecture built entirely on AWS managed services, organized into three processing layers: Collection, Analysis, and Delivery.

2.1 Architecture Overview

TRIGGER LAYER
Web Dashboard --> API Gateway --> Webhook Handler
EventBridge (06:00 UTC) -----> threat-collector

COLLECTION LAYER (VPC Private Subnet)
Lambda: threat-collector
-> AlienVault OTX + AbuseIPDB + VirusTotal
-> MITRE ATT&CK Mapping
-> DynamoDB (threat-iocs)

ANALYSIS LAYER (VPC Private Subnet)
Lambda: threat-analyzer
-> DynamoDB (read IOCs)
-> Amazon Bedrock Nova Lite (AI Analysis)
-> S3 (report.json)

DELIVERY LAYER (VPC Private Subnet)
Lambda: threat-reporter
-> Generate PDF (fpdf2)
-> AWS SES (Email + PDF)
-> Telegram Bot (notification)

2.2 AWS Services

Service	Purpose	Tier
AWS Lambda	Serverless compute for all 4 functions	Free Tier
Amazon DynamoDB	NoSQL storage for IOCs	Free Tier
Amazon S3	Storage for reports and web dashboard	Free Tier
API Gateway	HTTP endpoints for web and Telegram	Free Tier
Amazon Bedrock	AI analysis via Nova Lite model	Pay per use
AWS SES	Email delivery with PDF attachment	Pay per use
Secrets Manager	Secure storage of API keys	Pay per use
VPC + NAT Gateway	Network isolation for Lambdas	Pay per use
CloudWatch	Monitoring, logging, dashboards	Free Tier

3. Component Details

3.1 Lambda: threat-collector

Entry point of the pipeline. Fetches indicators from threat intelligence feeds, enriches critical IOCs with VirusTotal, maps them to MITRE ATT&CK techniques, and stores them in DynamoDB.

Processing Steps

- Fetch IOCs from AlienVault OTX (top 5 pulses x 10 indicators)
- Fetch malicious IPs from AbuseIPDB (confidence >= 90%, limit 50)
- Enrich top 10 critical IOCs via VirusTotal API
- Map each IOC to MITRE ATT&CK technique based on description keywords
- Batch write all IOCs to DynamoDB - async invoke threat-analyzer

3.2 Lambda: threat-analyzer

Reads today's IOCs from DynamoDB and sends a structured security summary to Amazon Nova Lite (Bedrock) for AI-powered threat analysis. Results are saved to S3 as a JSON report.

AI Analysis Output

```
{ "date": "2026-07-25", "total_iocs": 99, "critical_count": 50,
  "top_countries": [[ "RU", 25], [ "CN", 18]],
  "mitre_techniques": { "T1110": "Brute Force"},
  "analysis": "AI SOC report...", "analyzed_by": "Amazon Nova Lite" }
```

3.3 Lambda: threat-reporter

Generates a professional PDF report using fpdf2, sends it via email with AWS SES, and delivers a summary notification via Telegram.

PDF Report Sections

- Header with date and AI model information
- KPI cards: Total IOCs, Critical, High, VT Confirmed
- AI Security Analysis (full text from Nova Lite)
- Geographic Distribution bar chart (top 8 countries)
- IOC Severity Distribution + MITRE ATT&CK table

3.4 Lambda: threat-webhook-handler

Route	Method	Action
POST /webhook	POST	Process Telegram messages (/scan, /report, /status)
POST /scan	POST	Trigger pipeline from web dashboard
GET /report	GET	Return latest JSON report for dashboard
GET /status	GET	Return system status

OPTIONS /*	OPTIONS	CORS preflight response
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4. Security Architecture

4.1 Network Security (VPC)

All Lambda functions run inside a dedicated VPC (threat-intel-vpc) in a private subnet with no direct internet access. Internet connectivity is provided through a NAT Gateway.

Component	CIDR / Value	Purpose
VPC	10.0.0.0/16	Isolated network boundary
Public Subnet	10.0.1.0/24	NAT Gateway placement
Private Subnet	10.0.2.0/24	Lambda function isolation
NAT Gateway	Elastic IP	Outbound internet for private subnet

4.2 VPC Endpoints

Service	Type	Purpose
Amazon S3	Gateway	Private access to S3 (free)
Amazon DynamoDB	Gateway	Private access to DynamoDB (free)
AWS Secrets Manager	Interface	Secure secret retrieval
Amazon Bedrock Runtime	Interface	Private AI model invocation

4.3 Secrets Management (threat-intel/config)

Secret Key	Description
OTX_API_KEY	AlienVault OTX API key
ABUSEIPDB_API_KEY	AbuseIPDB API key
VIRUSTOTAL_API_KEY	VirusTotal API key
TELEGRAM_BOT_TOKEN	Telegram Bot API token
TELEGRAM_CHAT_ID	Default Telegram chat ID
S3_BUCKET	S3 bucket name for reports
SES_SENDER_EMAIL	Verified SES sender email
SES_RECIPIENT_EMAIL	Report recipient email

4.4 IAM Role: threat-lambda-role

Policy	Justification
AWSLambdaBasicExecutionRole	CloudWatch logging
AWSLambdaVPCAccessExecutionRole	VPC network interface creation
AmazonDynamoDBFullAccess	IOC read/write operations
AmazonS3FullAccess	Report storage and retrieval
AmazonBedrockFullAccess	Nova Lite model invocation
AmazonSESEFullAccess	Email report delivery
SecretsManagerReadWrite	Secret retrieval at runtime
AWSLambdaRole	Pipeline chaining between functions

5. Data Model

5.1 DynamoDB: threat-iocs

Attribute	Type	Description
ioc_value (PK)	String	The indicator value (IP, domain, URL, hash)
source (SK)	String	Data source (AlienVault / AbuseIPDB)
type	String	IOC type (IP, domain, URL, FileHash)
severity	String	CRITICAL or HIGH
country	String	Source country code (ISO 3166-1 alpha-2)
date	String	Collection timestamp (ISO 8601)
mitre_technique_id	String	e.g. T1110
mitre_technique_name	String	e.g. Brute Force
vt_malicious	Number	VirusTotal malicious engine count
vt_enriched	Boolean	Whether VT enrichment was performed

5.2 MITRE ATT&CK Keyword Mapping

Keyword	Technique ID	Technique Name
botnet	T1583.001	Acquire Infrastructure: Botnet
scanner	T1595	Active Scanning
bruteforce	T1110	Brute Force
malware	T1204	User Execution: Malicious File
phishing	T1566	Phishing
c2	T1071	Application Layer Protocol: C2
ransomware	T1486	Data Encrypted for Impact
tor	T1090.003	Proxy: Multi-hop Proxy
default	T1078	Valid Accounts

6. Threat Intelligence Sources

6.1 AlienVault OTX

Parameter	Value
API Endpoint	https://otx.alienvault.com/api/v1/pulses/subscribed
Authentication	X-OTX-API-KEY header
Pulses fetched	Top 5 subscribed pulses
Indicators per pulse	Top 10 per pulse
Cost	Free

6.2 AbuseIPDB

Parameter	Value
API Endpoint	https://api.abuseipdb.com/api/v2/blacklist
Confidence threshold	>= 90%
Limit per call	50 IPs
CRITICAL severity	Confidence >= 95%
HIGH severity	Confidence 90-94%
Cost	Free (1000 req/day)

6.3 VirusTotal Enrichment

Parameter	Value
Scope	Top 10 critical IOCs only
Data extracted	Malicious count, total engines, reputation
Rate limiting	0.5s delay between requests
Cost	Free tier (500 req/day)

7. Cost Analysis

Service	Usage	Monthly Cost
Lambda (4 functions)	30 executions/month	\$0.00 (Free Tier)
DynamoDB	~3,000 IOCs/month	\$0.00 (Free Tier)
S3 (2 buckets)	< 1GB total	\$0.00 (Free Tier)
API Gateway	< 1M requests/month	\$0.00 (Free Tier)
Amazon Bedrock Nova Lite	30 analyses x ~2K tokens	~\$0.01
AWS SES	30 emails/month	\$0.00 (Free Tier)
Secrets Manager	1 secret x 30 retrievals	~\$0.30
NAT Gateway	730 hours + data transfer	~\$32.00
VPC Endpoints (Interface x2)	2 x 730 hours	~\$14.00
CloudWatch	Basic metrics + logs	\$0.00 (Free Tier)
TOTAL		~\$46/month

Note: NAT Gateway and VPC Interface Endpoints represent ~99% of costs. For cost optimization, Lambdas can be placed in a public subnet, eliminating the need for NAT Gateway and Interface Endpoints.

8. Deployment Guide

8.1 Prerequisites

- AWS Account with credits or billing configured
- API keys: AlienVault OTX, AbuseIPDB, VirusTotal
- Telegram Bot token from @BotFather and Chat ID from @userinfobot
- Verified email address in AWS SES

8.2 Infrastructure Setup

- Create VPC (10.0.0.0/16) with public and private subnets
- Create Security Group: outbound HTTPS (443) only
- Create NAT Gateway in public subnet with Elastic IP
- Add route 0.0.0.0/0 -> NAT Gateway in private subnet route table
- Create VPC Endpoints: S3 Gateway, DynamoDB Gateway, Secrets Manager Interface, Bedrock Interface
- Create DynamoDB table: threat-iocs (PK: ioc_value, SK: source)
- Create S3 buckets for reports and web dashboard

8.3 Lambda Deployment

- Build fpdf2 Lambda Layer via CloudShell: `pip install fpdf2 --target python/ && zip -r layer.zip python/`
- Deploy all 4 Lambda functions with Python 3.12 runtime
- Attach all Lambdas to VPC private subnet with threat-intel-lambda-sg
- Set timeouts: threat-collector 5min, analyzer 3min, reporter 3min, webhook 5min
- Attach fpdf2 layer to threat-reporter

8.4 API Gateway & Web

- Create HTTP API with CORS enabled (Allow-Origin: *)
- Add routes: POST /webhook, POST /scan, GET /report, GET /status
- Set Telegram webhook URL to {API_URL}/webhook
- Update API URL in index.html and upload to S3 static website

9. Conclusion

The AI-Powered Threat Intelligence Platform demonstrates the practical application of cloud-native serverless architecture, artificial intelligence, and cybersecurity best practices. By combining AWS Lambda, Amazon Bedrock Nova Lite, and public threat intelligence feeds, the platform delivers enterprise-grade threat analysis at a fraction of the cost of commercial alternatives.

9.1 Key Achievements

- Full serverless architecture with zero server management overhead
- Multi-source threat intelligence with automated enrichment pipeline
- AI-powered analysis using Amazon Nova Lite (Bedrock)
- Enterprise security: VPC isolation, Secrets Manager, IAM least privilege
- Professional PDF generation and multi-channel delivery
- Interactive web dashboard hosted on AWS infrastructure

9.2 Future Enhancements

- EventBridge scheduler for automated daily scans at 06:00 UTC
- DynamoDB Streams for real-time critical IOC alerting
- Additional feeds: Shodan, PhishTank, URLhaus
- ML threat scoring model on Amazon SageMaker
- STIX/TAXII export format for SIEM integration
- Infrastructure as Code with Terraform